

Executive Summary — Heat-Driven Comfort & Water for Humid Climates

v1.4 — Two desiccant tracks, one physics, one platform · Land & marine

One sentence: Sun or waste heat in — dry cool air, safe air quality (<1,000 ppm CO₂), hot water, and drinking water out, for ~100 m³ occupied spaces (yacht interiors, small dwellings, cabins, shelters) in humid climates where no conventional passive or evaporative device can deliver comfort.

Status: Complete, internally consistent *paper design*, released as an open defensive publication (CERN-OHL-P v2 / CC-BY-4.0 / MIT). **Nothing has been built.** Every sizing claim is confidence-graded and gated on an explicit, deliberately cheap bench program: a few hundred dollars of tests for the liquid track; a mortar-and-pestle synthesis batch plus the M1–M4 coupon program for the solid track.

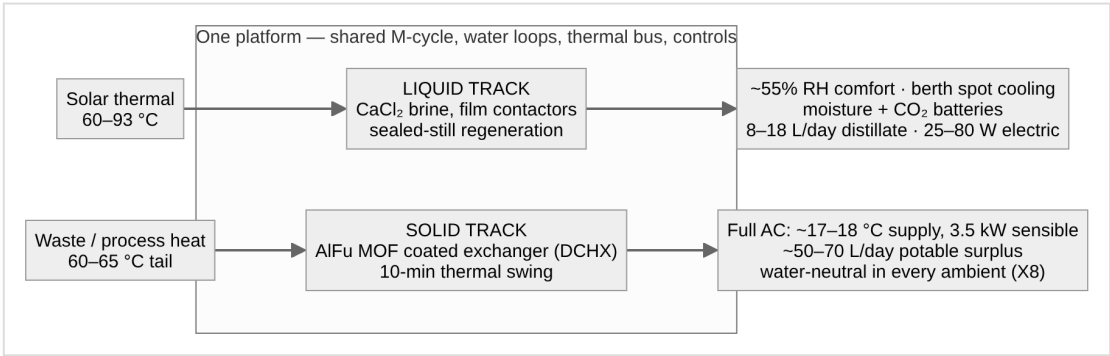
1. The problem — and why a desiccant is non-negotiable

The sole governing design point is **DP-A: 32 °C / 80% RH** (humidity ratio 24.2 g/kg, **dew point 28.1 °C**), with a raw-water sink (sea, lake, well) at **~29 °C**. At this point every ambient heat sink sits *at or above the dew point*, so:

- Sink-cooled condensation harvests essentially zero water from the air.
- Evaporative cooling — including the high-performance Maisotsenko (M-cycle) dew-point cooler — cannot cool below a 28 °C dew point on raw air.
- Vapor-compression AC works but demands 1–3 kW of electricity, a refrigerant circuit, and (marine) a salt-hostile machine.

Only a **desiccant** breaks the floor, because its surface vapor pressure is set by concentration or loading, not by a cold sink. The hard problem then becomes **regeneration against near-saturated surroundings** — never capture. That inversion of the published sorbent literature (which is tuned for arid harvesting) drives every design decision in the program.

2. The architecture — two complementary tracks



	Liquid track (CaCl ₂ brine)	Solid track (AlFu DCHX)
Role	Solar-grade layer — dehumidify to ~55% RH, spot cooling, storage batteries, degraded-mode backbone	Waste-heat layer — genuine full air-conditioning plus water surplus
Peak sorbent duty at DP-A	0.88 kg/h	~9–11 kg/h
Heat demand	~9–11 kWh/day at 60–93 °C — any source, incl. small solar	~7.5–9 kW continuous at 60–65 °C — waste-heat-coupled at peak
Electricity	25–80 W	~0.6–1.0 kW (pending measured ΔP)
Water	8–18+ L/day distillate	~50–70 L/day potable surplus
Storage	Moisture battery (~35–40 kg concentrate carries a full occupied night) + CO ₂ battery	Dry module as a small thermochemical store

Why two tracks is a theorem, not a hedge (finding X2): the liquid track regenerates in a *sealed still* with no purge stream — its driving force stays positive at any pool temperature above ~40 °C, so modest solar heat always works. The solid bed must desorb into a *condensing, humid purge* (~25 g/kg) and hits a zero-driving-force wall below ~50 °C (F2) — the widely quoted “~50 °C MOF regeneration” is a dry-purge artifact. Consequently **liquid = solar layer, solid = waste-heat layer**; they layer rather than compete, and both feed one M-cycle, one water system, one thermal bus.

3. Findings that changed the numbers (the honest core)

- **F1 — the dominant latent term was hidden.** At DP-A the 29 °C sink cannot absorb heat from a 25 °C cabin, so *all* cabin sensible load leaves evaporatively via M-cycle working air — whose moisture must pass through the desiccant. Full-AC duty is **~9–11 kg/h**, **not ~2 kg/h**; omitting this term under-sizes the desiccant 3–5×
- **X1 — once-through ventilation and whole-cabin M-cycle cooling never compose** (drying the required all-ambient supply costs 5.5–11 kg/h). Mixed-mode recirculation with a hard CO₂ interlock is the occupied baseline (X6).
- **X7 — an $\varepsilon \geq 0.8$ enthalpy-recovery ventilator (ERV)** cuts the fresh-air latent penalty ~9–10 kWh/day, making generous ventilation cheap.
- **X8 — saturated exhausts end in distillation recovery.** Closing the M-cycle working loop makes the cooler **water-neutral in every ambient** and seals the envelope against storm, spray, and salt aerosol.
- **X9/X10 — air quality is engineered, not assumed:** all-electric galley (no combustion in the envelope) and a two-bed solid-amine CO₂ battery — the only path holding <1,000 ppm sealed — regenerated by the same low-grade heat (~2 kWh/day).
- **The correction trail is kept visible** (nine liquid-track errata; F1's sharpening from 7–8 to 9–11 kg/h); *never cite a margin as a computed value; re-run every comfort claim when the design point moves; solve steady states simultaneously.*

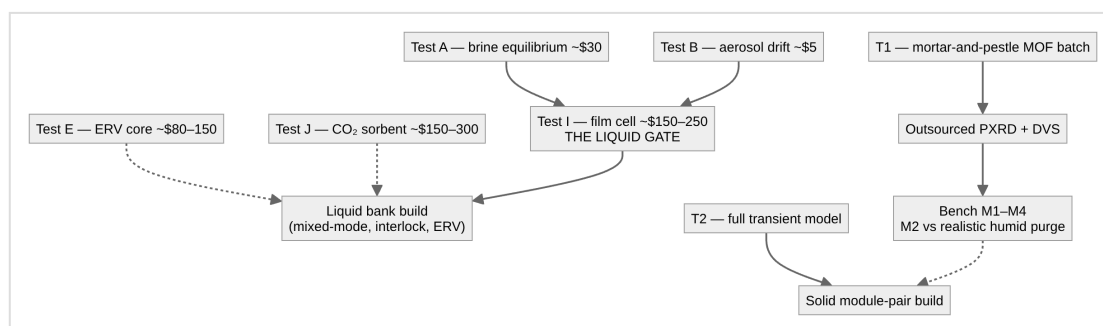
4. Platform integration

A heat-grade cascade organizes everything: high-grade process heat makes electricity and drives HDH desalination; its low-grade tail regenerates the solid track for free; solar thermal serves the liquid track. Water arrives by three *physically independent* routes — heat (HDH/ still), air (desiccant condensate), sky (rain) — with a pickled emergency RO unit as the only path independent of climate. **Governing reframe: redundancy outranks efficiency in the field.** On total loss of the primary heat source, dehumidified comfort and <1,000 ppm air quality survive on 2.5–4.5 m² of solar collector plus the moisture and CO₂ batteries.

5. Safety-critical requirements (binding)

CO₂ interlock (<1,000 ppm target / 2,000 alarm / per-room max sensing / **mechanical minimum stop** on the fresh damper) · no combustion appliances in the envelope · aerosol and amine-slip assays before any contactor or sorbent bed touches breathing air · independent potability and TDS tests before any water is drunk · crystallization interlock on the brine · the two-worlds chloride materials law on every wetted part. Any build omitting these departs from the design.

6. Validation path — cheapest decisive experiment first



The entire quantitative case is gated by bench-top spend: **a few hundred dollars** decides the liquid track end-to-end; the solid track's single most consequential measurement is **M2** — effective working capacity at 60–65 °C against a *representative humid purge* (a dry-purge test would merely reproduce the literature and decide nothing).

7. Bottom line

The physics closes on multiple independent validation passes. The design's value proposition is **resilience and heat flexibility, not watts**: near-silent, refrigerant-free comfort; freshwater as a *byproduct* instead of a cost; clean coupling to solar and waste heat; graceful degradation with engineered air-quality guarantees. A conventional vapor-compression AC remains the honest off-the-shelf benchmark — and the program says so. What stands between paper and proof is a few hundred dollars of deliberately sequenced bench tests.

Prepared from the repository document set 00–50, whose figures are collected with their confidence grades and gating tests in docs/parameter_register.xlsx.

Full enabling record: the complete document set, diagrams and parameter register are archived at [doi:10.5281/zenodo.21544099](https://doi.org/10.5281/zenodo.21544099) (concept DOI — always resolves to the current version), and developed openly at github.com/phycyclist/slice-cooling. This summary is an abstract; it is not itself enabling. Cite the DOI.

CPC classification (verified against the 2026.08 scheme; full titles in doc 50 §5): F24F 3/14 · F24F 3/1411 · F24F 3/1417 · F24F 3/1429 · F24F 3/147 · F24F 12/00 · F24F 5/0035 · B01D 53/14 · B01D 53/04 · B01D 53/26 · B01D 53/261 · B01D 53/28 · B01D 53/62 · B01D 53/1475 · C02F 1/04 · C02F 2103/08 · B63J 1/00 · B63J 2/02 · B63J 2/04. Not F24F 3/1423 — the solid track is a coated stationary exchanger, not a rotary wheel.

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Version history

- **v1.0** — Initial standalone abstract for the examiner channel (doc 50 §7), prepared from the v1.0 lineage (docs 00–40).
- **v1.1** — Brought up to the v1.2 lineage: provenance line now covers docs 00–50 (adding doc 31's upgrade paths and doc 50's disclosure procedure) and points at the parameter register; the bench-budget headline de-specified pending the doc 12 §4 reconciliation. No technical claim changed.
- **v1.2** — Footer gains the **concept DOI and repository URL**, which the document previously carried nowhere: doc 50 §7 deposits this file to the examiner channel precisely so it points searchers at the full enabling record, and it could not do that job. Provenance line corrected from lineage v1.2 to v1.4 — it had gone stale through v1.3, since this file was outside that release's bump set. No technical claim changed.
- **v1.3** — Carries the verified CPC classification codes. This file is the examiner-channel deposit (doc 50 §7), and examiners search classification before keywords; it previously carried none. No technical claim changed.
- **v1.4** — Provenance line no longer names a lineage version. It had gone stale twice in three releases (v1.2 → v1.3 → v1.4) because this file sits outside most bump sets and nothing checks a document's claim about the lineage's version. The version DOI already pins the exact set this summary accompanied, so naming it in prose adds a maintenance burden and a staleness risk while adding no information. No technical claim changed.